Phase Theory — Formal Axiomatization

Axiom Set Φ_0–Φ_{14} (Minimal Spine)

Dust LLC Research Team

Dust LLC · Phase Theory Research Group

Foundational Axioms — Core Formal System

May 2026

"Purpose: Provide a compact axiom set from which discreteness, causality, identity, locality, and emergent statistics follow as theorems."

Canonical source: *Phase Theory: A Unified Theory of Matter, Light, and Geometry* — Hanks (2026), Preprint.

Abstract

This paper presents the minimal axiomatic spine (Φ_0–Φ_{14}) of Phase Theory. Phase Theory proposes that phase is the sole fundamental constituent of physical reality, with space, time, matter, light, and geometry emerging from the global organization and topology of a single phase-bearing substrate. The axiom set introduced here formalizes the theory's foundational structure using six primitives — a phase configuration class, an extension relation, gauge equivalence, an admissibility predicate, and a compatibility predicate — from which discreteness, causality, identity, locality, measurement, and emergent statistics are derived as theorems rather than postulated. The axioms are stated in first-order logic with set-theoretic conventions. We enumerate the intended theorem pipeline and discuss minimality considerations.

Keywords: phase ontology, formal axiomatization, admissibility, topology, emergence, quantum foundations, unified theory

---

0 Preliminaries

0.1 Primitives

$\mathbf{\Phi}$: the Phase Theory admissibility formalism (not a field; the theory itself).

$\mathscr{P}$: a nonempty class of phase configurations.

$\sqsubseteq$: an extension relation on $\mathscr{P}$ (read: "is a restriction of / is extended by").

≃ : an equivalence relation on $\mathscr{P}$ (phase-gauge / representational equivalence).

Adm(·) : a predicate meaning "admissible" (defined axiomatically, not derived).

Comp(·,·) : a binary predicate meaning "compatible" (i.e., can be jointly realized).

Everything else (space, time, particles, probability, measurement) must be derived or defined.

0.2 Notational Conventions

- $p, q, r \in \mathscr{P}$
- $p \sqsubseteq q$ means q extends p.
- $[p]_{\simeq}$ is the ≃-equivalence class of p.

1 Axioms Φ_0–Φ_{14}

Φ_0 — Non-emptiness

$\mathscr{P} \neq \varnothing$

There exist phase configurations.

Φ_1 — Extension is a partial order

The relation $\sqsubseteq$ on $\mathscr{P}$ satisfies:

(i) Reflexive: $p \sqsubseteq p$

(ii) Antisymmetric: $p \sqsubseteq q \wedge q \sqsubseteq p \Rightarrow p = q$

(iii) Transitive: $p \sqsubseteq q \wedge q \sqsubseteq r \Rightarrow p \sqsubseteq r$

This is the structural substitute for "states evolving in time" — time is not assumed.

Φ_2 — Gauge equivalence

≃ is an equivalence relation and is extension-respecting:

If $p \simeq p'$ and $p \sqsubseteq q$, then $\exists q'$ such that $p' \sqsubseteq q'$ and $q \simeq q'$.

This prevents representational artifacts from becoming physics.

Φ_3 — Admissibility exists and is gauge-invariant

Adm is a primitive predicate satisfying:

Gauge invariance: $p \simeq q \Rightarrow (\mathrm{Adm}(p) \Leftrightarrow \mathrm{Adm}(q))$

Admissibility is the only "law source."

Φ_4 — Downward closure of admissibility

$\mathrm{Adm}(q) \wedge p \sqsubseteq q \Rightarrow \mathrm{Adm}(p)$

Any restriction of an admissible configuration is admissible.

Φ_5 — Non-factorizability (globality)

There is no decomposition of admissibility into purely local independent checks. Formally, there exists no family of predicates A_i such that for all p:

$$\mathrm{Adm}(p) \Leftrightarrow \wedge_i A_i(p|D_i)$$

for a fixed cover $\{D_i\}$ of the "support" of p (however support is later defined).

This axiom encodes the core "global consistency" property — admissibility is irreducibly global.

Φ_6 — Compatibility and join principle

Define compatibility: $\mathrm{Comp}(p, q)$ means "there exists r with $p \sqsubseteq r \wedge q \sqsubseteq r$ and $\mathrm{Adm}(r)$."

Axiom: If $\mathrm{Comp}(p, q)$, then $\exists r$ admissible such that:

- $p \sqsubseteq r, \quad q \sqsubseteq r$
- For any admissible s with $p \sqsubseteq s$ and $q \sqsubseteq s$, we have $r \sqsubseteq s$ (up to $\simeq$).

This is the structural substitute for "composition."

Φ_7 — Incompatibility is real (non-triviality)

There exist $p, q \in \mathscr{P}$ such that:

- $\mathrm{Adm}(p)$ and $\mathrm{Adm}(q)$
- but $\neg\mathrm{Comp}(p, q)$

This is the seed of contextuality, discreteness, and measurement selection.

Φ_8 — Extension determinacy constraint (no arbitrary branching)

For any admissible p, the set of admissible extensions of p is constrained:

$\mathrm{Adm}(p) \;\Rightarrow\; \exists \mathcal{E}(p) \subseteq \mathscr{P}$ such that $\forall q \;\; (p \sqsubseteq q \wedge \mathrm{Adm}(q) \Rightarrow q \in \mathcal{E}(p))$

and $\mathcal{E}(p)$ contains no mutually compatible distinct maximal elements (up to $\simeq$).

You don't get "free branching worlds" — maximal admissible completions are exclusive.

Φ_9 — Causal orientation as admissible extension

Define the causal precedence relation $p \prec q$ iff:

- $p \sqsubseteq q$
- and there is no admissible r with $p \sqsubseteq r \sqsubseteq q$ that reverses extension (i.e., no cycles).

Axiom: The admissible-extension graph is acyclic:

There is no finite sequence $p_0, \ldots, p_k$ with $k \geq 1$ such that:

- each p_i admissible,
- $p_i \sqsubseteq p_{i+1}$ for $i < k$,
- and $p_k \sqsubseteq p_0$.

This yields emergent causality without assuming time.

Φ_{10} — Identity as phase-continuity invariant (no copy-identity)

There exists an identity invariant mapping:

$$\mathscr{I} : \mathscr{P} \to \mathscr{S}$$

such that for admissible extension:

$\mathrm{Adm}(p) \wedge p \sqsubseteq q \;\Rightarrow\; \mathscr{I}(p) = \mathscr{I}(q)$

except at explicitly governed boundary transitions (introduced later in governance papers).

Crucially, structural similarity does not imply identity:

There exist p, q with $p \not\simeq q$ and $\mathscr{I}(p) \neq \mathscr{I}(q)$ even if some descriptive features match.

Φ_{11} — Locality as constrained influence (no controllable nonlocal rewriting)

There exists a notion of region restriction operator $R \mapsto p|R$ (defined within the formalism) such that:

(a) Restrictions respect extension: $p \sqsubseteq q \;\Rightarrow\; p|R \sqsubseteq q|R$

(b) No remote controllable rewriting: there does not exist an admissible mechanism by which changing $p|R_1$ forces an arbitrary choice of $p|R_2$ for spacelike-separated R_1, R_2 while preserving admissibility.

This is the formal "non-signaling" constraint — global correlation ≠ controllable remote influence.

Φ_{12} — Measurement as admissibility conditioning (no collapse postulate)

A "measurement context" is modeled as a constraint M that restricts admissible extensions:

Given admissible p, applying context M yields the admissible extension set:

$$\mathcal{E}_M(p) = \{q \in \mathcal{P} : p \sqsubseteq q \wedge \mathrm{Adm}(q) \wedge q \text{ satisfies } M\}$$

Axiom: Measurement never introduces new primitives; it only filters admissible continuations. There is no additional stochastic rule.

Φ_{13} — Emergent statistics from admissible-class measures

There exists a σ-algebra Σ over a suitable quotient of admissible completions (e.g., maximal admissible classes modulo ≃) and a measure μ such that:

- μ is invariant under gauge equivalence,
- Relative frequencies correspond to μ-weights of admissible classes under repeated trials / ensembles.

No intrinsic randomness is assumed — "probability" is emergent from admissible-class measure.

Φ_{14} — Closure and non-contradiction of the theory

The axioms are consistent and closed under their inference rules (standard first-order logic + any explicitly adopted rules), and the admissibility predicate never assigns 1 to a configuration that violates Φ_1–Φ_{13}.

This is the "global closure" statement — the system forbids internal self-contradiction by construction.

2 What This Axiom Set Is Designed to Yield (as Theorems)

- **Emergent discreteness** — from Φ_5–Φ_8 (incompatibility + non-factorizability + constrained maximal completions). Discreteness is not postulated but arises because admissibility is globally constrained and maximal completions are mutually exclusive.
- **Emergent causality** — from Φ_1, Φ_9 (acyclic admissible extension). The partial order on configurations, combined with the acyclicity axiom, yields a causal structure without assuming a background time parameter.
- **Identity continuity** — from Φ_{10} (no copy-identity). Identity is carried by phase-continuity through admissible extension, not by structural duplication or descriptive coincidence.
- **Non-signaling locality** — from Φ_{11} (correlation without remote control). Global admissibility permits nonlocal correlations, but no admissible mechanism allows controllable faster-than-light signaling.
- **Measurement without collapse** — from Φ_{12} (conditioning, not dynamics). Measurement is the restriction of the admissible extension set by a context constraint — no additional dynamical postulate or stochastic rule is required.
- **Statistics without intrinsic randomness** — from Φ_{13} (measure on admissible classes). Probabilistic predictions emerge from the gauge-invariant measure over equivalence classes of maximal admissible completions.

3 Minimality Notes

If you want the tightest spine, you can often compress to Φ_0–Φ_9 + Φ_{12}, treating identity/locality/statistics as definitional enrichments. But Φ_{10}–Φ_{13} make the "Phase-native replacement claim" explicit and non-optional.

4 References

[1] Hanks, M. (2026). *Phase Theory: A Unified Theory of Matter, Light, and Geometry.* Preprint. Under review.